2 Coffee Breaks — AM — 10:30 / PM — 3:00 > 15 mins

Lunch Break = 12:30 pm → 1 hr

Yes = +1 , No = +2.

"Github Copilot" = Chat Interface for coding
= Coding Buddy
↑
AI Coding Assistant = "AI, Assistant" ↓ Coding

(Generative AI Tool)

Generative AI → create a new content
(Text, code, Image, video)

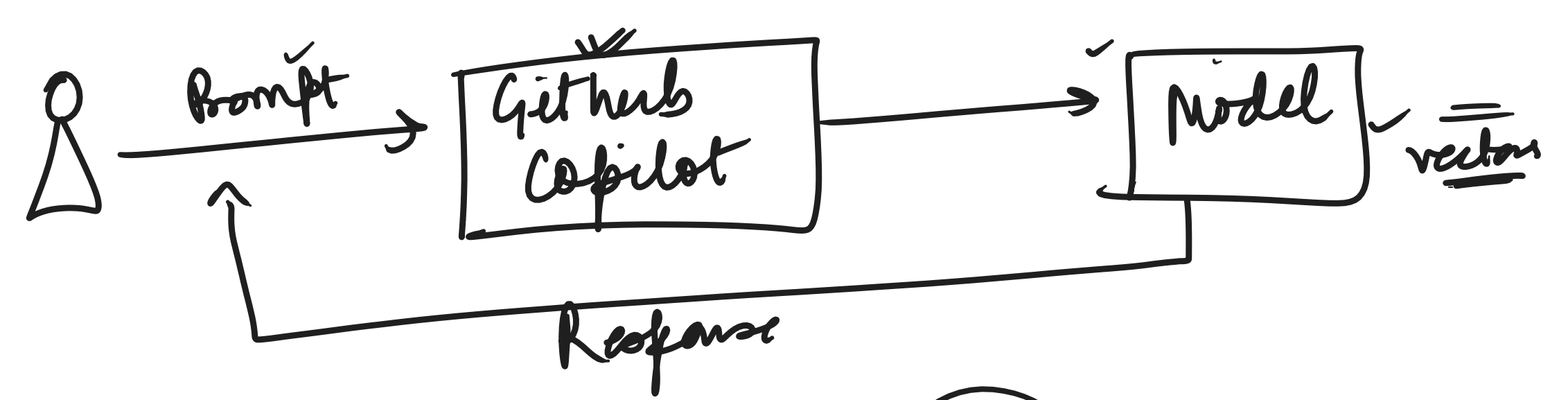
↳ Powered by FMs / LLMs

- ✓ FMs → Foundation Model
- ✓ LLMs → Large Language Model

] → General Domain Knowledge

Pretrained Model which are trained on Internet scale Data (Multi Modal Data)

Providers:
AI 21 → Jamba
Anthropic → Claude
Amazon → Nova, Titan
Google → Gemini
OpenAI → GPT



Model = Input
↳ Output = Tokens] (Tokens) = Input + output.
↑
Pricing
⇓
[Credits]

Prompt (Input) → 'Tokens'

(General) Model A | $0.007 (1M tokens) → Input + output

(Reasoning) B | $4.007 (1M tokens) Input , $8.007 (1M tokens) output.

C

Auto completions / Inline suggestions

- ✓ ① Inline suggestions / comments
- ✓ ② Inline chat ↵ → ctrl + i / cmd + i
- ✓ ③ Copilot chat

① Prompt Engineering for Copilot
② Best Practices + Prompt Patterns.
③ Debug-style prompting
④ Common Mistakes + fixes
⑤ Hands-on lab.

Copilot = "Context + Rules + Target output"
↳ Context = Existing code, file structure, examples.
↳ Rules / Constraints = language, style, lib, performance, security rules.
↳ Target output = What exactly we want produced
- function, Tests, Refactor, explanation, debugging steps.

PROMPT = "Build an API" ✗
↳ Copilot fills gaps with guesswork.

Golden Prompt Template

- ✓ 1. Goal = what you want
- ✓ 2. Inputs / Outputs = function signature, data shape
- ✓ 3. Rules = Must / must not
- ✓ 4. Edge cases : nulls, empty arrays, invalid input
- ✓ 5. Examples : one or two
- ✓ 6. Deliverables : code + tests + explanation

Best Practices:

① Give Copilot "the world it lives in":
- ✓ file
- ✓ sample data / nearby type

② Ask for incremental output, not a whole system

③ [Control style & safety]

Common Mistakes

✗ Vague prompts.
✗ Ask for too much at once
|API| + DB | + Auth + UI
✗ Copy code with verification.

Refactor the Code (Legacy code)

Rule: ① Ask Copilot to explain / plan
② Implement only one step
③ Run the app
④ Commit

Core use cases.

① Code Generation (functions, classes, APIs)
② Modernization (Callbacks → async / await, structure)
③ Documentation (docstring / JSDoc, Readme.md)
④ Unit Tests (Jest, SuperTest basics)
⑤ Integration (API + DB like in-memory store)
⑥ BPs, prompts pattern